



Service Bulletin Number: 4388840

Released Date: 12-nov-2015

Extended Service Interval Program for Mining Applications

Extended Service Interval Program for Mining Applications

Product Affected

- K38 (All Versions)
- K50 (All Versions)
- QSK23 CM500
- QSK23 CM2350 K127 MCRS
- QSK38 CM850 MCRS
- QSK38 CM2150 MCRS
- QSK38 CM2350 K125
- QSK45 CM500
- QSK50 CM2150 MCRS
- QSK50 CM2350 K108 MCRS
- QSK50 CM2350 K130 MCRS
- QSK50 CM850 MCRS
- QSK60 CM2150 MCRS
- QSK60 CM2350 K116 MCRS
- QSK60 CM2350 K121 MCRS
- QSK60 CM2350 K135 MCRS
- QSK60 CM2350 K136 MCRS
- QSK60 CM500
- QSK60 CM850 MCRS
- QSK78 CM2350 K126 MCRS
- QSK78 CM2350 K137 MCRS
- QSK78 CM2350 K154C MCRS
- QSK78 CM500
- QST30 CM2350 T101 MCRS

The following mine haul trucks are affected:

- Komatsu® 730E / 830E / 860E / 930E / 930-4SE / 960E / 980E / HD1500
- Hitachi® EH1100 / EH1700 / EH3500 / EH4000 / EH5000
- BelAZ® 75131 / 75139 / 75180 / 75305 / 75306 / 75307 / 75310 / 75603

The following excavators and shovels are affected:

- Hyundai™ R1200
- Komatsu® PC3000 / PC4000 / PC5500 / PC7000 / PC8000
- Hitachi® EX1200 / EX1900 / EX2500 / EX2600 / EX3600 / EX5500 / EX5600 / EX8000

Note : If an engine model or an application is not listed above, contact Cummins Distributor or Cummins Filtration

Content Overview

The purpose of this document is to lay out the steps required to extend the service interval for off-highway mining applications.

Section 1: Site Assessment

Section 2: Fluid Specifications

Section 3: Fluid and Filter Monitoring

Section 4: Initial Equipment Upgrades and Filter Requirements

Section 5: Extended Maintenance Intervals

Section 6: Lubricating Oil Drain Interval Extension Test

Section 7: Additional Service Literature

Maintenance intervals discussed within this document include:

- Air filters
- Coolant drain
- Coolant filters
- Fuel filters
- Fuel tank vent filters
- Lubricating oil drain
- Lubricating oil filters

Note : The guidelines and requirements in this bulletin are subject to change without notice. Consult your local Cummins Distributor or Cummins Filtration.

Note : Before adopting this information, it is the responsibility of the end user to contact Cummins Distributor or Cummins Filtration to understand the information in its entirety. Cummins Inc. is not responsible for inaccurate handling of this procedure and failure to Cummins Warranty.

Extended Service Interval for Off-highway Mining Applications Program Requirements

Section 1: Site Assessment

The equipment, site, operating conditions, and environment affect the ability to extend maintenance intervals. The make and model of the equipment, site's rigorous maintenance practice, higher and low loads, long idling periods, and weather and environmental extreme conditions such as wet, dusty, freezing, and hot operation might require more frequent maintenance than specified in the Maintenance Manual.

A site assessment **must** be completed to ensure success with this extended maintenance interval program. To schedule a site assessment, contact your local Cummins Distributor or Cummins Filtration.

Section 2: Fluid Specifications

Note : Any deviations from the fluid (lubricating oil, fuel, coolant) and filter specifications as defined in this service bulletin will void the use of this extended service bulletin. Fluid and filters **not** meeting the requirements listed in this document **must** follow the standard maintenance intervals defined in the corresponding Operation and Maintenance Manual.

Lubricating Oil Requirements

- Lubricating oil **must** meet the requirements specified by Cummins Inc.
 - See Fluids for Cummins Products Service Manual, Bulletin 5411406 (</qs3/pubsys2/xml/en/bulletin/5411406.html>).
- Lubricating oil used **must** be registered in Cummins Oil Registration List.

Note :

https://quickserve.cummins.com/qs3/qsol/service/serviceproducts/oil_registration.html
(https://quickserve.cummins.com/qs3/qsol/service/serviceproducts/oil_registration.html)

- Lubricating oil drain intervals **must** follow the intervals specified in the corresponding Operation and Maintenance Manual unless extended drain intervals are proven through a testing protocol described in this bulletin.
 - For best results, use a fully synthetic oil certified to CES 20078 or CES 20086.
 - Some blended oils or natural base oils can achieve extended service intervals.
- If sulfur content in fuel is higher than 15 parts per million (ppm), oil drain intervals **must** be verified using the procedure in Section 6.
- If sulfur content in fuel is higher than 500 ppm, extended drain intervals will **not** be supported.

Note : During testing it is best **not** to mix two types or brands of oil, even if both are registered to an appropriate Cummins specification. In some cases, it has been observed that mixing oils led to challenges in interpreting the effects of extended oil drains from the mixing of oils.

Fuel Requirements

- Fuels **must** meet the requirements as specified by Cummins Inc.
 - See Fluids for Cummins® Products Service Manual, Bulletin 5411406 (</qs3/pubsys2/xml/en/bulletin/5411406.html>).

Specific fuel requirements include:

- Fuels **must** meet International Organization for Standardization (ISO) 4406 cleanliness code of 18/16/13 or better at the inlet of the stage 1 fuel filter with a maximum water content of 200 ppm.
- Fuels that meet or exceed American Society for Testing and Materials (ASTM) D2068 "Filter Plugging Tendency Tests" procedure A with a Filter Blocking Tendency (FBT) calculation high limit of 1.41.
- Fuel must pass the fuel quality test using the Cummins Fleetguard Portable Fuel Cleanliness Analysis Kit, part number FK36000.
- Fuels that meet or exceed ASTM D975 - 15b "Standard Specification for Diesel Fuel Oils."
- **Only** fuel additives that have demonstrated no negative effect on fuel filter or fuel system life can be used.
- If winter, cold weather, or high altitude fuel is used:
 - The applicability of this service bulletin is limited to fuels **not** exceeding 60% kerosene / 40% diesel content.
- If biofuel is used:
 - The applicability of this service bulletin is limited to fuels **not** exceeding B5 or 5% biofuel content.
- If sulfur content in fuel is higher than 15 ppm, lubricating oil drain intervals **must** be verified using the procedure in Section 6.
- If sulfur content in fuel is higher than 500 ppm, extended drain intervals will **not** be supported.

Coolant Requirements

Note : Do not mix two types of coolant products or brands.

Note : Do not top off coolant system with water or conventional coolant.

- Coolant **must** meet the requirements specified by Cummins Inc.
 - See Fluids for Cummins® Products Service Manual, Bulletin 5411406 (</qs3/pubsys2/xml/en/bulletin/5411406.html>)
- Coolants used **must** be registered in Cummins Coolant Registration List and in accordance with CES 14603 coolant specification and be fully formulated pre-mixed organic acid technology (OAT) coolant. Reference Coolant Registration List.

Note :

https://quickserve.cummins.com/qs3/qsol/service/serviceproducts/coolant_registration.html
(https://quickserve.cummins.com/qs3/qsol/service/serviceproducts/coolant_registration.html)

- The following coolants have been proven capable of meeting extended drain intervals:

- Fleetguard® ES Compleat™ EG OAT Coolant
- Fleetguard® ES Compleat™ PG OAT Coolant
- Fleetguard® PG Platinum™ Heavy Duty Engine Coolant
- Cummins® TEC Infinity™ Extended Life Coolant
- Cummins® PGXL™ Vital Advanced Extended Life Heavy Duty Coolant

Section 3: Fluid and Filter Monitoring

This section provides the guidelines and requirements to collect, monitor, flag, act, and analyze engine fluids and filters that provide information to guide the Extended Service Interval for Off-highway Mining Applications Program.

Lubricating Oil Sampling and Analysis:

- See Service Bulletin, Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060 (/qs3/pubsys2/xml/en/bulletin/4022060.html).
 - Elements to Monitor for High Horse Power Engine Oil Analysis
 - Oil Sample Collection
 - Setting Oil Analysis Flag Limits
 - Investigation and Corrective Action Based on Flag Limits
- See Service Bulletin, Statistical Analysis of Oil Sample Lead Readings on High Horsepower Engines, Bulletin 2883452 (/qs3/pubsys2/xml/en/bulletin/2883452.html).

Fuel Sampling and Analysis:

- For High Pressure Injection (HPI) Engine, Cummins Inc. recommends fuel samples be taken directly from vehicle/equipment at the inlet to suction side fuel filter.
- For Modular Common Rail System (MCRS) Engines, samples **must** refer to Bulletin 4022123 (/qs3/pubsys2/xml/en/bulletin/4022123.html), Fuel Sampling for Particle Counting Test.
- If the fuel sample does **not** meet the requirements specified, revisit the site audit to identify opportunities for improved fuel quality.

Coolant Sampling Analysis:

- See the corresponding Operation and Maintenance Manual, under Section V - Maintenance and Specification, procedure 018-004 Coolant Recommendations and Specifications.
- If molybdate, nitrite, and pH are **not** within specification, investigate maintenance practices and send coolant sample for contamination check. Coolant **must** be replaced. See the corresponding Service Manual. Reference procedure 008-018 Cooling System.
- If freeze point is **not** within specification when measuring with 4-way additive/freeze point strips, Fleetguard® Part Number CC8997, verify the results with a refractometer. If freeze point is confirmed to be out of specification, replace coolant. See the corresponding Service Manual. Reference procedure 008-018 Cooling System.

Table 1 and Table 2 below specify the fluid analysis to be performed during operation when using extended service intervals.

Table 1, Engine Fluid Sampling Analysis Requirements

Fluid	Sampling Interval	Suggested Method and Location	Purpose and Test Package
Lubricating Oil	250 hours	Per Service Bulletin4022060 (/qs3/pubsys2/xml/en/bulletin/4022060.html): Collect oil sample from pressurized port while engine idling or by vacuum pump through lubricating oil dipstick retaining tube.	Test ¹ for Oil Quality and Contaminants: • Elemental Metals by Inductively Coupled Plasma (ICP) • Viscosity at 100°C [212°F] • Water Percentage • Fuel Dilution • Soot Percentage • Oxidation / Nitration • Total Base Number (TBN)
Coolant	4000 hours	By vacuum pump through radiator fill cap or through quick fill / drain port	Test ² for Coolant Replacement Limits: • Sulfate • Chloride, Fluoride, Bromide • pH • Oil and fuel contaminants • Grease, solder bloom, silica gel, rust or scale
Coolant	Every 6 months	By vacuum pump through radiator fill cap or through quick fill / drain port	Test for Glycol Levels: • Coolant Test Kit, Fleetguard Part Number CC8997
Fuel	As Needed: Sample fuel from equipment for quality and cleanliness if fuel filters do not reach required life or fuel filter life decreases from established baseline.	Use sampling port upstream of Stage 1 fuel filters and at bottom of equipment fuel tank through port on equipment fuel tank.	Reference the Fuel Filter Plugging Bulletin, 5504538 (/qs3/pubsys2/xml/en/bulletin/5504538.html).
If using Cummins® Fleetguard® Monitor: ¹ Reference Monitor™ CC36136, ² Reference Monitor™ CC2700.			

Table 2, Bulk Storage, Tank Storage, and Service Equipment Fluid Sampling Requirements

Fluid	Sampling Interval	Suggested Method and Location	Purpose and Tests Cummins® Monitor Program Test Package
Bulk Storage – Lubricating Oil	Minimum of quarterly and if change to lubricating oil brand or specification	By sampling valve installed in bulk tank	Test ¹ for Oil Quality and Contaminants: • Elemental Metals by Inductively Coupled Plasma (ICP) • Viscosity at 100°C [212°F] • Water Percent • Fuel Dilution • Soot Percent • Oxidation / Nitration • Total Base Number (TBN)
Bulk Storage – Coolant	Minimum of quarterly and if change to coolant brand or specification	By sampling valve installed in bulk tank	Test ² for Coolant Quality and Contaminants: • Visuals (foam, color, oil, fuel, magnetic and non- magnetic precipitate and odor) • 12 Metals by Inductively Coupled Plasma (ICP) • Percent Antifreeze • Freeze Point • Boil Point • pH • Hardness • SCA Number • Nitrites • Total Dissolved Solids • Specific Conductance

Table 2, Bulk Storage, Tank Storage, and Service Equipment Fluid Sampling Requirements

Fluid	Sampling Interval	Suggested Method and Location	Purpose and Tests Cummins® Monitor Program Test Package
Bulk, Tank Storage and Field Service Equipment - Fuel	Monthly or more frequently if fuel filters of equipment do not reach required life or filter life decreases from established baseline	By sampling valve installed in bulk tank, with particle counter in place and sample bottles. At bottom of equipment fuel tank through port on equipment fuel tank with sample bottles	Test ³ for Fuel Quality and Contaminants: • 24 Metals by Inductively Coupled Plasma (ICP) • Viscosity at 40°C [104°F] • Percent Sulfur • API Gravity • Distillation • Flash Point • Cloud Point • Pour Point • Water and Sediment • Bacteria, Fungi and Mold • Calculated Cetane Index • Thermal Stability • Water by Karl Fischer titration method
			Test ⁴ for Fuel ISO Particle Counts and Contaminants: • PPM ISO Particle Count
			Test for Filter Plugging: • Cummins Fleetguard Portable Fuel Cleanliness Analysis Kit, part number FK36000 or ASTM D2068 - Procedure A.
If using Cummins® Fleetguard® Monitor™: ¹ Reference Monitor™ CC36136, ² Reference Monitor™ CC2700, ³ Reference Monitor™ CC2651, ⁴ Reference Monitor™ CC2719			

For more information about Fleetguard® Monitor™, reference CumminsFiltration.com.

Section 4: Initial Equipment Upgrades and Ongoing Filter Requirements

Note : The guidelines and requirements in this bulletin are subject to change without notice. Consult your local Cummins Distributor or Cummins Filtration.

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Note : Failure that might occur due to installing or using other manufacturer's auxiliary devices, accessories or consumables, are not covered under Cummins® Warranty.

Note : Any deviations from the fluid and filter specifications as defined in this service bulletin will void the use of this extended service bulletin. Fluid and filters not meeting the requirements listed in this document must follow the standard maintenance intervals defined in the corresponding Operation and Maintenance Manual.

Extending maintenance intervals requires the following:

- Fuel: Advanced filtration inclusive Industrial Pro™ fuel systems with Nanonet® media
- Lubricating Oil: Venturi™ Combo lube filters and Spiratec™ centrifuge lubricating oil filtration
 - Spin-on filters: Venturi™ Combo, Stratapore™, Nononet™ Combo, and/or
 - Eliminator: Spiratec™ centrifuge lubricating oil filtration
- Air Filtration: Nanoforce® air filtration
- Coolant: Stratapore™ extended service coolant filtration

Lubricating Oil Filters

For engines with spin-on lubricating oil filters, add remote centrifuge and use a one-piece centrifugal oil filter with SpiraTec™ as shown in Tables 3 and 4.

Note : Reference the installation instructions included in the box with the remote centrifuge.

For engines with Eliminator™, replace Eliminator™ rotor with one-piece centrifugal lubricating oil filter with SpiraTec™ technology as shown in Table 5. Reference Technical Service Bulletin 160036 (/qs3/pubsys2/xml/en/bulletin/160036.html).

- Perform the following checks:
 1. The casting tolerance of the centrifuge cover has been adjusted to accommodate the new centrifugal filter. Centrifuge covers with the adjusted tolerance are stamped with the letter "V". If the centrifuge cover is **not** stamped with the letter "V", replace the centrifuge cover with a centrifuge cover, Part Number 4016423, stamped with the letter "V" or grind/file any high spots found at the parting line on the inside of the centrifuge cover.
 2. Measure clearance between the spindle and the bushings inside the one-piece centrifugal lubricating oil filter, replace the spindle if out of specification. Reference Bulletin 3666253 (/qs3/pubsys2/xml/en/bulletin/3666253.html).

Table 3, On-Engine Lubricating Oil Filter Part Numbers

Original Filtration Part Numbers				Upgraded Filtration Part Numbers Fleetguard® Part Number		
Engine Model	Description	Cummins® Part Number	Fleetguard® Part Number	Description	Cummins® Part Number	Fleetguard® Part Number

Table 3, On-Engine Lubricating Oil Filter Part Numbers

	Original Filtration Part Numbers			Upgraded Filtration Part Numbers Fleetguard® Part Number		
K38 (All Versions) K50 (All Versions) QSK23 CM500	On-Engine Spin-On Lubricating Oil Filter	3310169	LF3325	Venturi™ Combo-spin-on filters with StrataPore™	4096196	LF9325
QSK38 CM850 / CM2150 QSK50 CM850 / CM2150 QSK60 CM500 / CM850 / CM2150 QSK78 CM500	Venturi™ Combo spin-on filters with StrataPore™	4920071	LF9050	On-Engine Full Flow filters with Nanonet Media	5375337	LF14005NN
QSK38 CM2350 / QSK50 CM2350	Venturi™ Combo spin-on filters with StrataPore™	3647559	LF9540	On-Engine Spin-On Lubricating oil filter	3648053	LF17564
QSK60 CM2350	On-Engine Spin-On Lubricating Oil Filter	3648053	LF17564	Not Applicable	Not Applicable	Not Applicable
QSK78 C2350	On-Engine Spin-On Lubricating Oil Filter	4349561	LF17562	Not Applicable	Not Applicable	Not Applicable

Table 4, Remote Centrifuge Housing Part Numbers

Engine Model	Description	Cummins® Part Number	Fleetguard® Part Number
QSK23 CM500	Remote Centrifuge	Not Applicable	CH41111
K38 (All Versions) QSK38 (All Versions) K50 (All Versions)	Remote Centrifuge – side drain	5373170	CH44120
QSK50 (All Versions) QSK60 (All Versions) QSK78 (All Versions)	Remote Centrifuge – bottom drain	5373171	CH44105

Table 5, Centrifugal Oil Filter Part Numbers for Eliminator™ or Remote Centrifuge

Engine Model	Description	Cummins® Part Number	Fleetguard® Part Number
QSK23 CM500	One-Piece Centrifugal Oil Filter with SpiraTec™ Technology	4968791	CS41011
K38 (All Versions) QSK38 (All Versions) K50 (All Versions) QSK50 (All Versions) QSK60 (All Versions)	One-Piece Centrifugal Oil Filter with SpiraTec™ Technology	4331734	CS41018
QSK78 (All Versions)	One-Piece Centrifugal Oil Filter with SpiraTec™ Technology	3649569	CS41019

Fuel Filters

For engines with a High Pressure Injection (HPI)® fuel system, remove the on-engine fuel filter head and install off engine IndustriaPro fuel filters. Reference installation instructions included in the box with the fuel filters.

For engines with a Modular Common Rail System (MCRS) fuel system, upgrade stage 1 filtration to IndustriaPro fuel filters. Reference installation instructions included in the box with the filters.

Table 6, Fuel Filter Head and Fuel Filter Element Part Numbers

Engine Model	Filtration Stage	Description	Quantity	Cummins Part Number	Fleetguard Part Number	Quantity	Cummins Part Number	Fleetguard Part Number
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Table 6, Fuel Filter Head and Fuel Filter Element Part Numbers

K38 (All Versions) QSK38 (All Versions) QSK23 CM500	Stage 1	Industrial Pro Unit, Single, Tall, Bracket, WIF, Clear Bowl with 5 Micron NanoNet ® filter. In Left / Out Right	1	4328042	FH23901	1	4328043	FS53014
		Industrial Pro Unit, Single, Tall, Bracket, WIF, Clear Bowl with 5 Micron NanoNet ® filter. In Right / Out Left	1	4328041	FH23903	1	4328043	FS53014
		Industrial Pro Unit, Single, Tall, Bracket, WIF, Clear Bowl with 5 Micron NanoNet ® filter. In Left / Out Left	1	4328064	FH23905	1	4328043	FS53014

Table 6, Fuel Filter Head and Fuel Filter Element Part Numbers

		Industrial Pro Unit, Single, Tall, Bracket, WIF, Clear Bowl with 5 Micron NanoNet ® filter. In Right / Out Right	1	4328065	FH23906	1	4328043	FS53014
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Table 6, Fuel Filter Head and Fuel Filter Element Part Numbers

K50 (All Versions) QSK50 (All Versions) QSK60 (All Versions)	Stage 1	Industrial Pro Unit, Dual Tall, Bracket, WIF, Clear Bowl with 5 Micron Nanonet filter. In Right / Out Right	1	4381962	FH23907	2	4328043	FS53014
		Industrial Pro Unit, Dual Tall, Bracket, WIF, Clear Bowl with 5 Micron Nanonet filter. In Right / Out Left	1	4381959	FH23908	2	4328043	FS53014
		Industrial Pro Unit, Dual Tall, Bracket, WIF, Clear Bowl with 5 Micron Nanonet filter. In Left / Out Right	1	4381960	FH23910	2	4328043	FS53014

Table 6, Fuel Filter Head and Fuel Filter Element Part Numbers

		Industrial Pro Unit, Dual Tall, Bracket, WIF, Clear Bowl with 5 Micron Nanonet filter. In Left / Out Left	1	4381961	FH23911	2	4328043	FS53014
QSK78 (All Versions)	Stage 1	Industrial Pro Unit, Triple Tall, Bracket, WIF, Clear Bowl with 5 Micron Nanonet filter. In Left / Out Right	1	5374777	FH23930	3	4328043	FS53014
QSK38 CM850 CM2150 CM2350 ¹	Stage 2	On Engine Configuration Filter Head with 3 ² Micron NanoNet ® filter	1	4380732	Not Applicable	2	5375819	FF63042 NN
QSK50 CM850 CM2150 CM2350 ¹	Stage 2	On Engine Configuration Filter Head with 3 ² Micron NanoNet ® filter	1	4380732	Not Applicable	2	5375819	FF63042 NN

Table 6, Fuel Filter Head and Fuel Filter Element Part Numbers

QSK50 CM850 CM2150 CM2350	Stage 2	Remote Mount Configuration ² Filter Head with 3 Micron NanoNet® filter	1	4380859	Not Applicable	3	5375819	FF63042 NN
QSK60 CM850 CM2150 CM2350 ¹	Stage 2	On Engine OR Remote Mount Configuration Filter Head with 3 Micron NanoNet® filter	1	4380859	Not Applicable	3	5375819	FF63042 NN
QSK78 CM2350 ¹	Stage 2	On Engine OR Remote Mount Configuration Filter Head with 3 Micron NanoNet® filter	1	4380859	Not Applicable	3	5375819	FF63042 NN

¹Fuel filter head upgrade is not required on QSK50, QSK60, and QSK78 CM2350 engines.

²Excavators and shovels using QSK50 engines with on engine stage 2 fuel filters (2 filter elements) may not reach extended maintenance intervals. Switch to remote mount filter head (3 filter elements) for this application to achieve extended maintenance intervals.

Fuel Filters

For engines with a High Pressure Injection (HPI)® fuel system, remove the on-engine fuel filter head and install off engine IndustriaPro fuel filters. Reference installation instructions included in the box with the fuel filters.

For engines with a Modular Common Rail System (MCRS) fuel system (except CM2350), upgrade stage 1 filtration to IndustriaPro fuel filters. Reference installation instructions included in the box with the filters.

Table 7, Associated Hardware

Description	Cummins® Part Number
2 Stage Air Bleed Valve	4379736

Note : Air bleed valve is only needed when upgrading fuel filter head. Fuel filter head upgrade is not required on QSK50, QSK60 and QSK78 CM2350 engines.

Fuel Tank Vent Filters

The equipment fuel tank is to have a filtration system fitted to the fuel tank vent to remove contaminants from the ingested air. A fuel tank vent filter prevents fine airborne particles from contaminating the fuel tank as air is drawn in. The fuel tank venting system **must** accommodate the fastest rate of fuel fill that the vehicle will experience. See equipment manufacturer service information.

- For equipment with existing fuel tank vent filters, install upgraded fuel tank vent filter as defined in Table 8 below. See equipment manufacturer service information for installation information.
- For equipment without existing fuel tank vent filters, install both a fuel tank vent system as defined in Table 9 below and the fuel tank vent filter as defined in Table 8 below. See equipment manufacturer service information for installation information.

Table 8, Fuel Tank Vent Filter Part Number

Description	Fleetguard® Part Number
Fuel tank vent filter, 3 micron	HF30195

Table 9, Fuel Tank Vent System Part Number

Description	Hydrau-Flo® Part Number
Fuel tank vent system suitable for 1000 liters per minute fast fill	HF1008Z0010

Coolant Filters

The upgraded coolant filtration options offer higher capacity and improved filtration performance. Remove existing coolant filters and replace with StrataPore™ coolant filters. See Table 10 below for part numbers.

Table 10, Coolant Filter Part Numbers

Description	Cummins® Part Number	Fleetguard® Part Number
On-engine, no DCA*, StrataPore™ media	3098690	WF2123
*Only filters with no DCA should be used in combination with OAT coolant.		

Air Filters



WARNING

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

For all engines, replace existing primary air filters with upgraded air filters. See Table 11 below for part numbers.

For engines, without air filter restriction indicators installed, install air filter restriction indicators. For information on the installation of air filter restriction indicators, contact equipment manufacturer.

To reach extended maintenance intervals, maximum allowable intake restriction with clean filter element is 253 mm H₂O [10 in H₂O].

Table 11, Air Filter Part Numbers

Type	Description	Original Fleetguard® Part Numbers	Upgrade NanoForce® Fleetguard® Part Numbers
Axial Seal	Primary	AF1605 or AF1605M	AF1605NF
	Secondary	AF1604	No change in part number
Axial Seal	Primary	AF879 or AF879M	AF879NF
	Secondary	AF857	No change in part number
Axial Seal	Primary	AF899 or AF899M	AF899NF
	Secondary	AF880	No change in part number
Radial Seal	Primary	AF27696	AF27993NF + SP1307 service cover*
	Secondary	AF27695	AF27994
Radial Seal	Primary	AF27698	AF25149NF + SP1307 service cover*
	Secondary	AF27697	AF25150

*One time installation of SP1307 is required to install AF27993NF

Section 5: Extended Maintenance Intervals

⚠ CAUTION ⚠

Never wash or clean air filters. Filter washing or cleaning may compromise the air filter media and engine damage can result.

Note : Refer to the Service Manual for different ambient temperature engine start up conditions. Refer to Fluid Cummins Product for change of elements on different ambient conditions.

Seasonal Requirements:

Cold Weather Preparation:

- Clean air housings. See equipment manufacturer service information.
- Check operation of fuel filter heaters.

Table 12, Time Based Maintenance Intervals for Haul Trucks

Syste m	List of Maint enanc e Tasks	As Requi red	Daily and Weekl y	250 Hours	500 Hours	1000 Hours	1500 Hours	2000 Hours	3000 Hours	4000 Hours	5000 Hours
Coola nt	Chan ge coola nt filters									X	
Coola nt	Top off coola nt	X			X	X	X	X	X	X	X
Fuel	Drain water and sedim ent from lower bowl of Indust rial Pro™ fuel filters	X	X	X	X	X	X	X	X	X	X
Fuel	Repla ce stage 1 fuel filters	X				X		X	X	X	X
Fuel	Repla ce stage 2 fuel filters	X				X		X	X	X	X
Fuel	Repla ce fuel tank breath er filters	X				X		X	X	X	X

Table 12, Time Based Maintenance Intervals for Haul Trucks

Syste m	List of Maint enanc e Tasks	As Requi red	Daily and Weekl y	250 Hours	500 Hours	1000 Hours	1500 Hours	2000 Hours	3000 Hours	4000 Hours	5000 Hours
Lubricating Oil	Collect lubricating oil sample and perform lubricating oil analysis	X		X	X	X	X	X	X	X	X
Lubricating Oil	Change engine lubricating oil		According to Operation and Maintenance Manual or Lubricating Oil Drain Interval Established in Section 6.								
Lubricating Oil	Change rotor in Eliminator™ / Centrifuge					X		X	X	X	X
Lubricating Oil	Change lubricating oil filters (if installed)					X		X	X	X	X

Table 13, Time Based Maintenance Intervals for Excavators and Loaders

System	List of Main tenance Task s	As Requ ired	Daily and Week ly	250 Hour s	500 Hour s	1000 Hour s	1500 Hour s	2000 Hour s	3000 Hour s	4000 Hour s	5000 Hour s
Air	Chan ge prim ary air filter elem ents	X			X	X	X	X	X	X	X
Air	Chan ge seco ndary air filter elem ents	X			X	X	X		X		
Air	Inspe ct and clean air filter housi ngs	X									X
Coolant	Test coola nt for repla ceme nt limits	X								X	
Coolant	Chan ge coola nt filters									X	
Coolant	Top off coola nt	X			X	X	X	X	X	X	X

Table 13, Time Based Maintenance Intervals for Excavators and Loaders

System	List of Main tenance Task s	As Requ ired	Daily and Week ly	250 Hour s	500 Hour s	1000 Hour s	1500 Hour s	2000 Hour s	3000 Hour s	4000 Hour s	5000 Hour s
Fuel	Drain water and sedi ment from lower bowl of Indus trial Pro™ fuel filters	X	X	X	X	X	X	X	X	X	X
Fuel	Repla ce stage 1 fuel filters	X			X	X	X	X	X	X	X
Fuel	Repla ce stage 2 fuel filters	X			X	X	X	X	X	X	X
Fuel	Repla ce fuel tank breat her filters	X			X	X	X	X	X	X	X

Table 13, Time Based Maintenance Intervals for Excavators and Loaders

[illegible]

Table 13, Time Based Maintenance Intervals for Excavators and Loaders

System	List of Main tenance Task s	As Requ ired	Daily and Week ly	250 Hour s	500 Hour s	1000 Hour s	1500 Hour s	2000 Hour s	3000 Hour s	4000 Hour s	5000 Hour s
Lubricating Oil	Chan ge rotor in Elimi nator TM / Centr ifuge					X	X	X	X	X	X
Lubricating Oil	Chan ge lubric ating oil filters (if instal led)					X	X	X	X	X	X

Table 14, Condition Based Filter Maintenance Intervals

System	Replacement Indication
Fuel Filters, Stage 1	Replace based on schedule in Tables 12 and 13 or using visual indication on Industrial Pro TM units or when indicated by FleetguardFIT TM .
Fuel Filters, Stage 2	Replace based on schedule in Tables 12 and 13 or when indicated by FleetguardFIT TM .
Fuel Tank, Vent / Breather Filters	Replace based on schedule in Tables 12 and 13.
Lubricating Oil Filters	Replace based on schedule in Tables 12 and 13 or when indicated by FleetguardFIT TM .
Lubricating Oil, Eliminator TM / remote centrifuge rotor	Replace based on schedule in Tables 12 and 13 or at the same time as lubricating oil and lubricating oil filters are replaced.
Coolant / Water filters	Replace based on schedule in Tables 12 and 13.
Air filter, Primary	Replace based on schedule in Tables 12 and 13 or when the differential reaches 635 mm-h ₂ O [25 in- h ₂ O] on the restriction indicator, whichever occurs first or when indicated by FleetguardFIT TM .
Air filter, Secondary	Replace every third primary air filter change.

FleetguardFIT™

Filtration Intelligence Technology—is a real-time filtration monitoring system that provides full visibility to filter (Fuel, Lube and Air) and oil life enabling proactive problem detection using intelligent sensing and advanced data analytics. To learn more about FleetguardFIT and how to get it onto your engine, contact Cummins Filtration at info@fleetguardfit.com.

Note : If fuel filters do not reach required life or filter life decreases, assessment of fuel quality, fuel tank condition, storage systems and bulk filtration system is required.

Section 6: Lubricating Oil Drain Interval Extension Test

⚠ CAUTION ⚠

The use of a synthetic base lubricating oil does not justify extended lubricating oil change intervals. Extended oil change intervals can decrease engine life as the result of corrosion, deposits, and wear.

⚠ CAUTION ⚠

If the equipment is using a remote lubricating oil tank in which lubricating oil is continuously circulated (not a Centinel™ oil make-up tank), the lubricating oil in the remote tank must be changed when the lubricating oil in the engine lubricating oil sump is changed.

The purpose of this test is to increase the interval between lubricating oil changes using lubricating oil analysis results from the Fleetguard® Monitor Fluid Analysis Program or another equivalent laboratory.

This method for setting the Lubricating Oil Drain Interval with Filter Optimization replaces the use of the Lubricating Oil Chart Method when using this Service Bulletin.

Do **not** extend lubricating oil and lubricating oil filter change intervals beyond the limits specified in the maintenance schedule, unless all of the conditions required by this service bulletin are met:

- Completion of site assessment as defined in Section 1.
- Lubricating oil meets Specifications in Section 2.
- Use of lubricating oil filters specified in Section 3.
- Lubricating oil monitoring system is in place as specified in Section 4.

If fleet is made up of mixed engine models, varying duty cycles, or equipment with differing lubricating oil system configurations, each group **must** be considered separately. A minimum of three units of each should be evaluated thoroughly prior to adopting a drain interval for the fleet. Segment engines by:

- Engine Model
- Lubricating oil system capacity (including any remote lubricating oil tanks where the lubricating oil is continuously circulated)
- Use of Centinel™ lubricating oil replenishment systems
- Use of Eliminator™ vs spin on lubricating oil filters and centrifuge
- Duty cycle
- Lubricating oil used (brand name, performance category, and viscosity grade)

Note : The Lubricating Oil Drain Interval Extension established by this process can **only** be applied to engines in the same segment as defined above and with the specific lubricating oil used to establish the extended lubricating oil drain interval. Any changes to the lubricating oil (including changes to viscosity grade), duty cycle, or engine lubricating oil system will require re-verification of the lubricating oil drain interval.

Lubricating oil samples **must** be sent to a lab capable of testing the parameters in Table 14 using the specified test method.

Table 15, Oil Analysis Test Parameters

Parameter Type	Parameter, Unit	Recommended Test Method
Lubricating Oil Health	Total Base Number (TBN), mg KOH/g	ASTM D4739
	Total Acid Number (TAN), mg KOH/g	ASTM D664
	Oxidation, abs/cm	ASTM E2412 or ASTM D7414*
	Nitration, abs/cm	ASTM E2412
	Viscosity at 100°C [212°F], cSt	ASTM D445
Contamination	Fuel Dilution, massPercent fuel in oil	ASTM D7593
	Soot, massPercent soot in oil	ASTM E1131
	Sodium - Na, ppm	D5185
	Potassium - K, ppm	D5185
	Silicon - Si, ppm	D5185
	Aluminum - Al, ppm	D5185
Wear Metals	Iron - Fe, ppm	D5185
	Lead - Pb, ppm	D5185
	Copper - Cu, ppm	D5185
	Aluminum - Al, ppm	D5185
	Tin - Sn, ppm	D5185
	Chromium - Cr, ppm	D5185
*The Fourier transform infrared spectroscopy (FTIR) data for oxidation can use either ASTM method but should report whether trending or subtraction methods were used.		

Collection of Lubricating Oil Sample:

Samples **must** be taken at the intervals specified in Table 17 or 18 below. For sampling procedure, See Service Bulletin, Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060 ([/qs3/pubs3/engines2/xml/en/bulletin/4022060.html](https://pubs3/engines2/xml/en/bulletin/4022060.html)).

A minimum of five lubricating oil samples per oil drain interval **must** be collected to establish trend lines. For lubricating oil that can not reach the interval established in the sampling plan, collect lubricating oil samples at more frequent intervals (every 50 hours or every 75 hours) to determine the appropriate lubricating oil drain interval.

Label the sample with the following information before submitting for analysis:

- Engine model/serial number
- Miles/hours of lubricating oil use
- Miles/hours accumulated on engine since new or rebuild
- Lubricating oil used (brand name, performance category and viscosity grade)
- Date sample was collected
- Engine application
- Amount of new lubricating oil added since previous lubricating oil change.

Verification of Lubricating Oil Integrity:

Both trending and limits are used to determine the appropriate lubricating oil drain interval. Cummins Inc. and the Customer should collaborate to determine the appropriate lubricating oil drain interval. Documentation of the lubricating oil drain interval validation **must** be retained in the site assessment file.

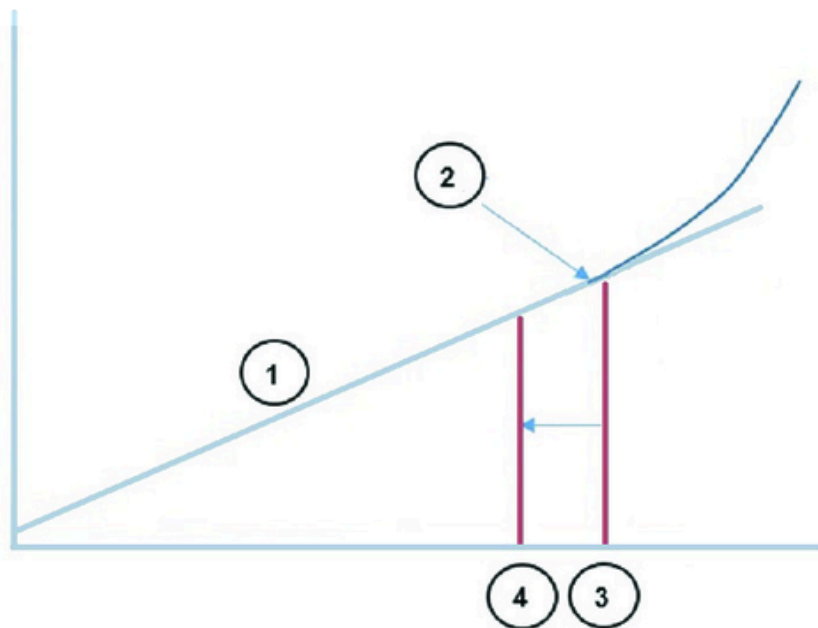
1. Linear trending throughout each lubricating oil drain interval of:

- Total Base Number (TBN)
- Total Acid Number (TAN)
- Oxidation
- Nitration
- Soot (Soot may **not** begin trending immediately; when trending soot is usually linear.)
- Iron
- Lead
- Copper

Linear Trending Example Key for Figure 1 Below:

1. Linear trend of lubricating oil analysis parameter value vs hours on lubricating oil.
2. Point at which lubricating oil analysis parameter value deviates from linear trend line.
3. Hours on lubricating oil at which lubricating oil analysis parameter value deviates from linear trend line.
4. Established lubricating oil drain limit. Subtract 20% from hours on lubricating oil where parameter value deviates from linear trend line.

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Figure 1, Linear Trending Example

2. Lubricating Oil Analysis Parameters are within limits defined in Table 16 below.

Table 16, Oil Analysis Parameter Limits

Oil Analysis Parameter	Limit Value
Total Acid Number (TAN) mg KOH/g	2.5 increase over new lubricating oil or equal to TBN
Total Base Number (TBN) mg KOH/g	2.5 minimum or equal to TAN

Table 16, Oil Analysis Parameter Limits	
Oil Analysis Parameter	Limit Value
Fuel Dilution Percent	5%
Soot Percent	5%
Viscosity 100°C [212°F] cSt	For XW-40:
	MIN: 12.5 MAX: 16.3
	For XW-30:
	MIN: 9.3 MAX: 12.5
Oxidation abs/cm – direct trending	15 ¹
Nitration abs/cm – direct trending	25
¹ Increase over new (unused oil sample)	

3. Repeat sample plan for 3 different lubricating oil drain intervals on each of the 3 engines in the same segment.

4. Final lubricating oil drain will be the hours at which any of the samples did **not** meet the lubricating oil integrity test criteria reduced by 20%.

- a. 20% margin is added to the approved lubricating oil drain interval to account for changes in rate of lubricating oil degradation as the engine ages.

Table 17, Example Lubricating Oil Sampling Plan for Haul Trucks		
Sample Number	Lubricating Oil Hours	Description
0	Baseline	Sample unused lubricating oil from the container
1	0	Collect sample after filling the sump with new lubricating oil and letting the engine idle for 5 minutes
2	250	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.
3	500	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.
4	600	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.
5	700	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.

Table 17, Example Lubricating Oil Sampling Plan for Haul Trucks		
Sample Number	Lubricating Oil Hours	Description
6	800	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.
7	900	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.
8	1000	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.
9	1050	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.
10	1100	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.
11	1150	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.
12	1200	End of test. Collect sample before draining lubricating oil.

Table 18, Example Lubricating Oil Sampling Plan for Excavators and Shovels		
Sample Number	Lubricating Oil Hours	Description
0	Baseline	Sample unused lubricating oil from container
1	0	Collect sample after filling sump with new lubricating oil and letting engine idle for 5 minutes
2	200	Verify Lubricating Oil Integrity using steps above. Stop test and change lubricating oil if requirements are not met.
3	300	Verify Lubricating Oil Integrity using steps above. Stop test and change lubricating oil if requirements are not met.

Table 18, Example Lubricating Oil Sampling Plan for Excavators and Shovels

Sample Number	Lubricating Oil Hours	Description
4	400	Verify Lubricating Oil Integrity using steps above. Stop test and change lubricating oil if requirements are not met.
5	500	Verify Lubricating Oil Integrity using steps above. Stop test and change lubricating oil if requirements are not met.
6	550	Verify Lubricating Oil Integrity using steps above. Stop test and change lubricating oil if requirements are not met.
7	600	End of test. Collect sample before draining lubricating oil.

Section 7: Additional Service Literature

Literature found on QuickServe® Online:

Beginning in 2007, Cummins Inc. started using service model names. For assistance in identifying associated service model name:

- See Technical Service Bulletin, Cummins® Service Engine Model Identification, TSB130080. Review associated Cummins® Product Technology procedures.

Service Bulletins:

- See Service Bulletin, Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060 (/qs3/pubsys2/xml/en/bulletin/4022060.html).
- See Service Bulletin, Statistical Analysis of Oil Sample Lead Readings on High Horsepower Engines, Bulletin 2883452 (/qs3/pubsys2/xml/en/bulletin/2883452.html).
- See Service Bulletin, Fuel Sampling for Particle Counting Test, Bulletin 4022123 (/qs3/pubsys2/xml/en/bulletin/4022123.html).

Operation and Maintenance Manuals:

- See Cummins® Service Engine Model Identification, Bulletin TSB130080. Review the associated Cummins® Product Technology procedures.
- See QSK45 and QSK60 (MCRS and HPI) Operation and Maintenance, Bulletin 3666260.
- See K38, K50, QSK38 and QSK50 Operation and Maintenance Manual, Bulletin 3810497.
- See QSK38 CM2350 K125 Operation and Maintenance Manual, Bulletin 5613390.
- See QSK45 and QSK60 Operation and Maintenance Manual, Bulletin 4915528.
- See QSK50 CM2350 K108 (MCRS) Operation and Maintenance Manual, Bulletin 4332824.
- See QSK60 CM2350 K116 (MCRS) Operation and Maintenance Manual, Bulletin 4367445.
- See QSK60 CM2350 K121 (MCRS) Operation and Maintenance Manual, Bulletin 4388646.
- See QSK60 CM2350 K135 (MCRS) Operation and Maintenance Manual, Bulletin 5504625.
- See QSK60 CM2350 K136 (MCRS) Operation and Maintenance Manual, Bulletin 5504630.
- See QSK78 CM2350 K126 (MCRS) Operation and Maintenance Manual, Bulletin 5504265.
- See QSK78 CM2350 K137 (MCRS) Operation and Maintenance Manual, Bulletin 5504635.
- See QSK78 CM2350 K154C (MCRS) Operation and Maintenance Manual, Bulletin 5676943.
- See QST30 CM2350 T101 (MCRS) Operation and Maintenance Manual, Bulletin 5467180.

- See QSK23 CM2350 K127 (MCRS) Operation and Maintenance Manual, Bulletin 5504159.
- See QSK50 CM2350 K130 (MCRS) Operation and Maintenance Manual, Bulletin 5504179.

Service Manuals:

- Fluids for Cummins Products Service Manual, Bulletin 5411406.
- See Cummins® Service Engine Model Identification, Bulletin TSB130080. Review the associated Cummins® Product Technology procedures.
- See QSK45 and QSK60 (MCRS and HPI) Service Manual, Bulletin 4021530.
- See K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528.
- See QSK38 Service Manual, Bulletin 5613391.
- See QSK45 and QSK60 Service Manual, Bulletin 4021530.
- See QSK50 CM2350 K108 Service Manual, Bulletin 4332823.
- See QSK60 CM2350 K116 Service Manual, Bulletin 4367444.
- See QSK60 CM2350 K121 Service Manual, Bulletin 438 (/qs3/pubsys2/xml/en/bulletin/4388645.html)8645 (/qs3/pubsys2/xml/en/bulletin/4388645.html).
- See QSK60 CM2350 K135 Service Manual, Bulletin 5504626.
- See QSK60 CM2350 K136 Service Manual, Bulletin 5504631.
- See QSK78 CM2350 K126 Service Manual, Bulletin 5504266.
- See QSK78 CM2350 K137 Service Manual, Bulletin 5504636.
- See QSK78 CM2350 K154C Service Manual, Bulletin 5676944.
- See QST30 CM2350 T101 Service Manual, Bulletin 5467179.
- See QSK23 CM2350 K127 Service Manual, Bulletin 5504160.
- See QSK50 CM2350 K130 Service Manual, Bulletin, Bulletin 5504180.

Literature **not** found on QuickServe® Online:

Hard Copy Service Bulletin:

- Service Bulletin, Fuel Quality for High Horsepower Engines with Modular Common Rail Fuel Systems (MCRS), Bulletin 4091849 (/qs3/pubsys2/xml/en/bulletin/4091849.html).

Document History

Date	Details
2015-11-12	Module Created
2016-5-12	Added information about the lubricating oil filters for the K50. Added Tables 1 and 5. Updated Test for Filter Plugging in Tables 10 and 11.
2016-8-30	Added checks to the Lubricating Oil Filters section. Updated Cummins® part numbers in Table 2.
2016-12-1	Added QSK78 Engine.
2017-10-24	Updated engines in product affected. Added excavators and shovels to Product Affected. Modified Table 1, 2, 3, 4, and 12. Added Table 13 and 18.
2018-6-12	Updated QSK78 Spiratec part number, 2 fuel filter part numbers, fuel testing recommendations, and coolant testing recommendations.
2020-2-20	Update part numbers, engine models and document structure.
2023-6-13	Updated Product Affected section.

Date	Details
2023-7-11	Updated product affected section and tables.
2023-7-14	Corrected minor formatting error.
2026-6-18	Update Service Bulletin 4388840 Product Affected section to include QSK38 CM2350 K125 mining application engines used in Komatsu PC3000 and PC5500 excavator applications already covered by the bulletin.

Last Modified: 18-Jun-2026
